LABS 1–11: Probability and Statistics Lab

Diabetes and lifestyle indicators

Statistical analysis with R
BMAT202P Probability and Statistics Lab

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Data → graphs → models → statistical tests

Dataset and aim

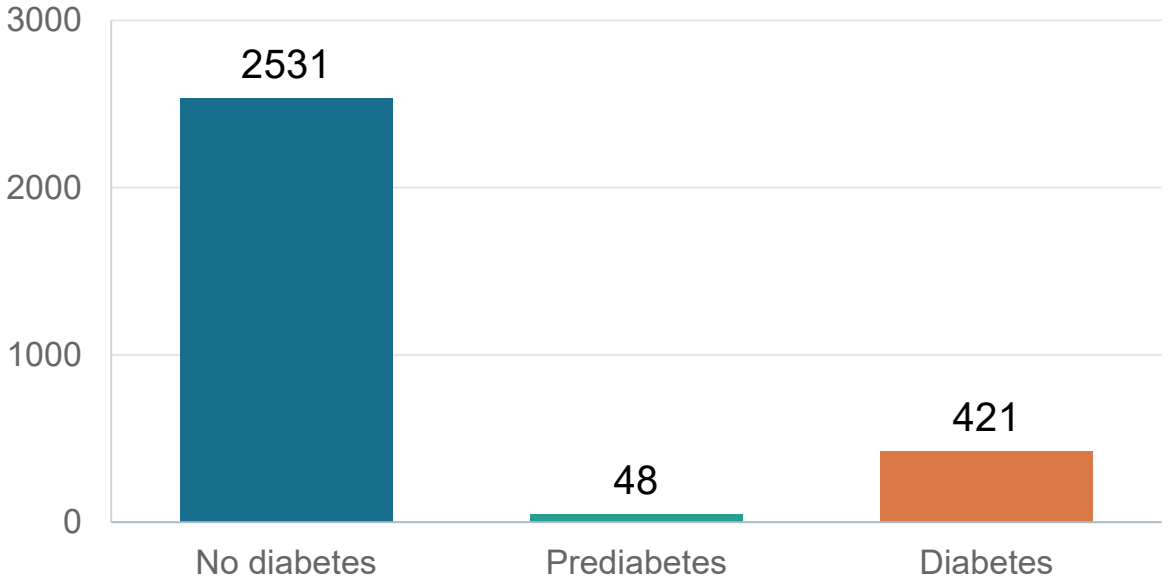
Aim: describe health patterns, study BMI relationships and apply the course statistical methods.

253,680
source records

3,000
sampled people

21
health predictors

Health status in the sample



Key variables

Variable	What it measures
BMI	Body mass index (kg/m²)
Mental / physical health	Poor-health days, 0–30
Age / income / education	Grouped categories
Activity / smoking	Yes / No indicators

Source: CDC BRFSS 2015, processed by Alex Teboul on Kaggle.

469 people (15.63%) report prediabetes or diabetes.

Data preparation

Import → sample → validate → prepare for analysis

Import
CSV source

Sample
3,000 random rows

Validate
Codes and ranges

Prepare
Labels and types

Types used in R

Type	Examples
Numeric	BMI and poor-health days
Categorical	Health status, sex and smoking
Grouped categories	Age, income and education

Quality checks

Check	Result
Missing entries	0
Invalid codes / ranges	0
Health-day range	0–30 days
Sample selection	Seed 101

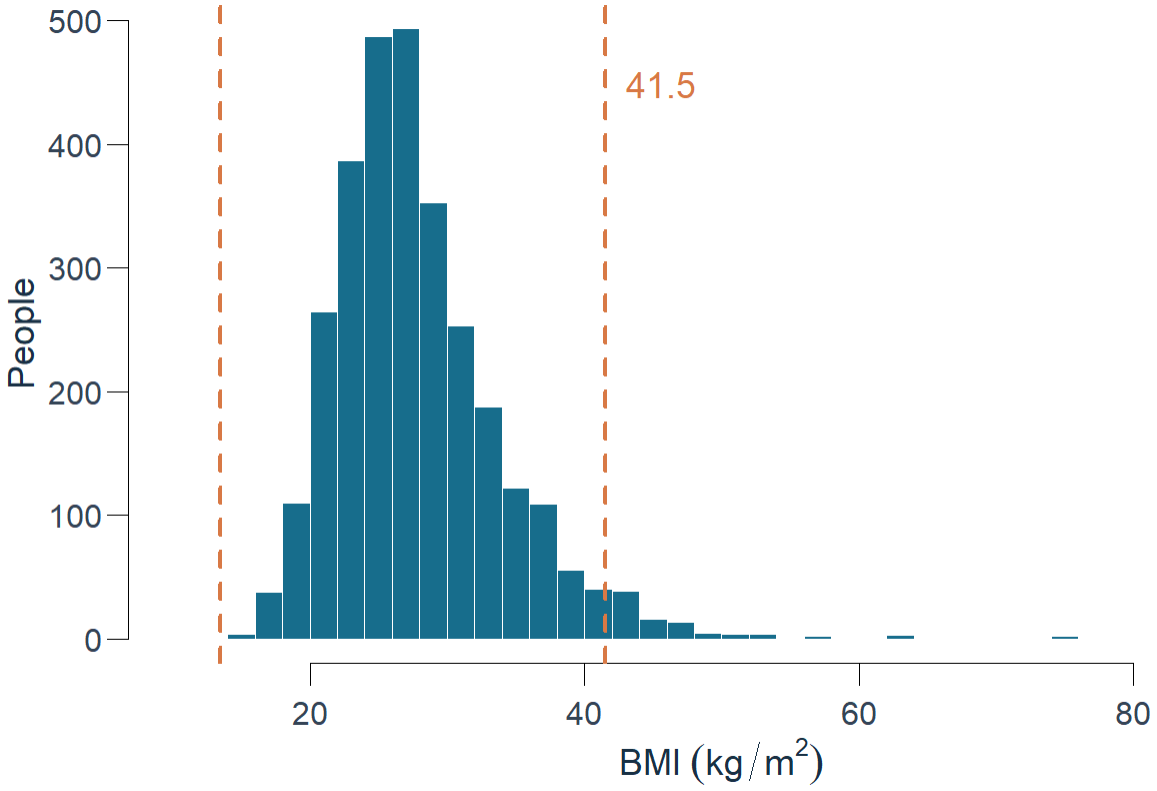
Import CSV → label categories → export clean data for the exercises

Readable labels and correct data types make every later analysis easier.

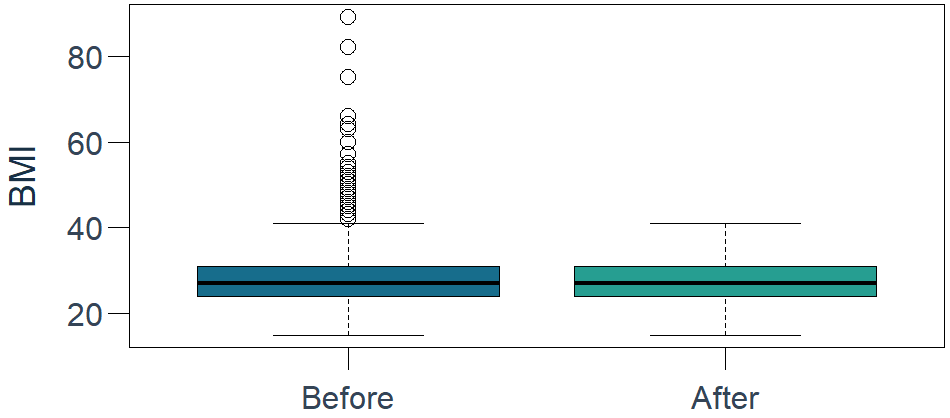
Outlier analysis and removal

Q1 = 24 • Q3 = 31 • IQR = 7 • Q1 − 1.5×IQR to Q3 + 1.5×IQR → 13.5 to 41.5

Detect unusual BMI values



Compare before and after



Measure	Before	After
People	3,000	2,886
Mean BMI	28.40	27.65
BMI variance	40.34	24.59
Positive status	15.63%	14.62%

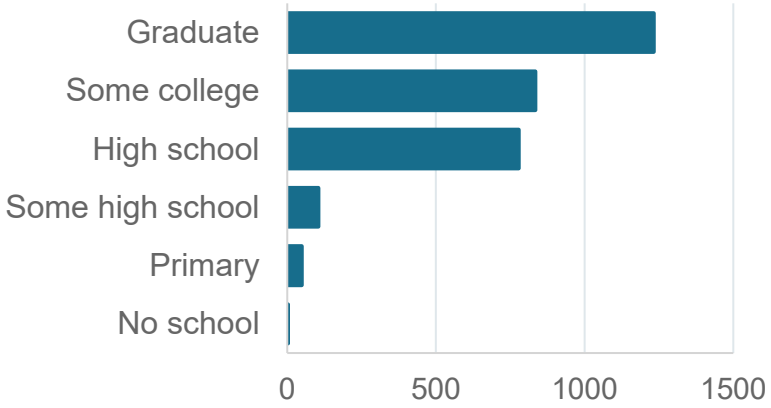
114 high BMI values (3.8%) were flagged. Removing them lowers the average and spread.

Summary statistics and graphs

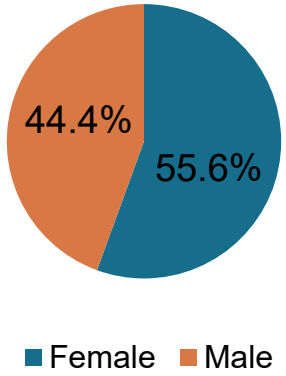
Mean = average • SD = standard deviation • Variance = squared spread

Variable	Mean	Median	SD	Variance
BMI	28.40	27	6.35	40.34
Mental-health days	3.26	0	7.39	54.67
Physical-health days	4.15	0	8.62	74.24

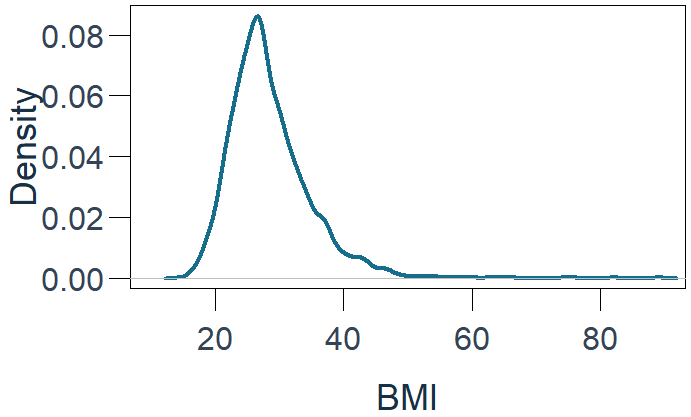
Education: group sizes



Sex: sample share



BMI: distribution shape

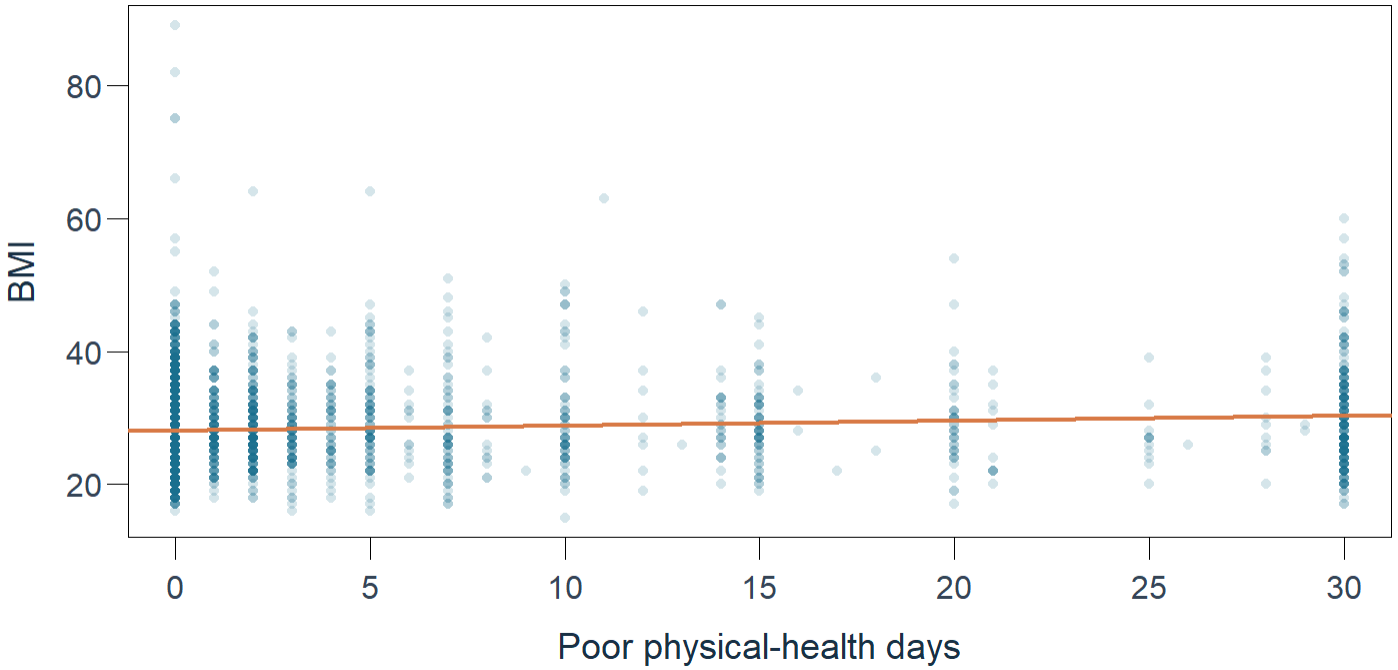


BMI has a long right tail. Most people report few poor-health days.

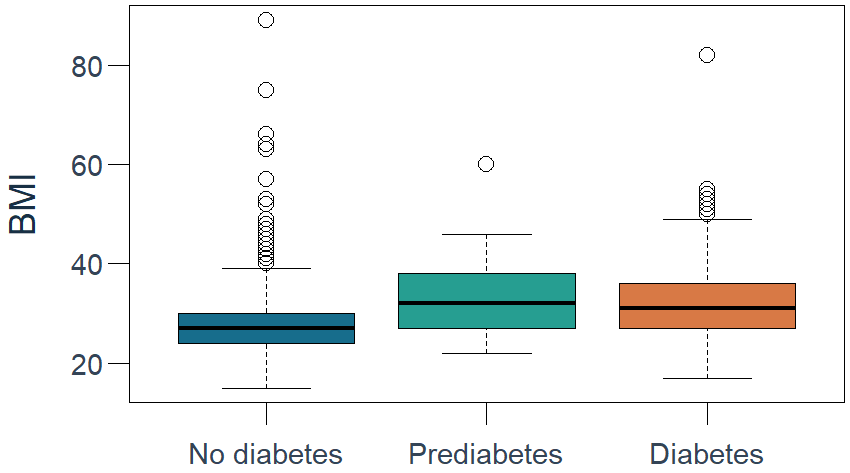
Correlation and simple regression

Predicted BMI = 28.09 + 0.0749 × poor physical-health days

Poor-health days and BMI



BMI by health group



Pearson r	Spearman r	Covariance	R^2
0.102	0.108	5.56	1.03%

Average BMI

No diabetes	Prediabetes	Diabetes
27.72	32.85	31.98

r = 0.102: weak positive relation. R^2 = 1.03%: very little BMI variation is explained.

Multiple regression and model checks

Predict BMI using age, activity, income, education and smoking. R^2 = variation explained.

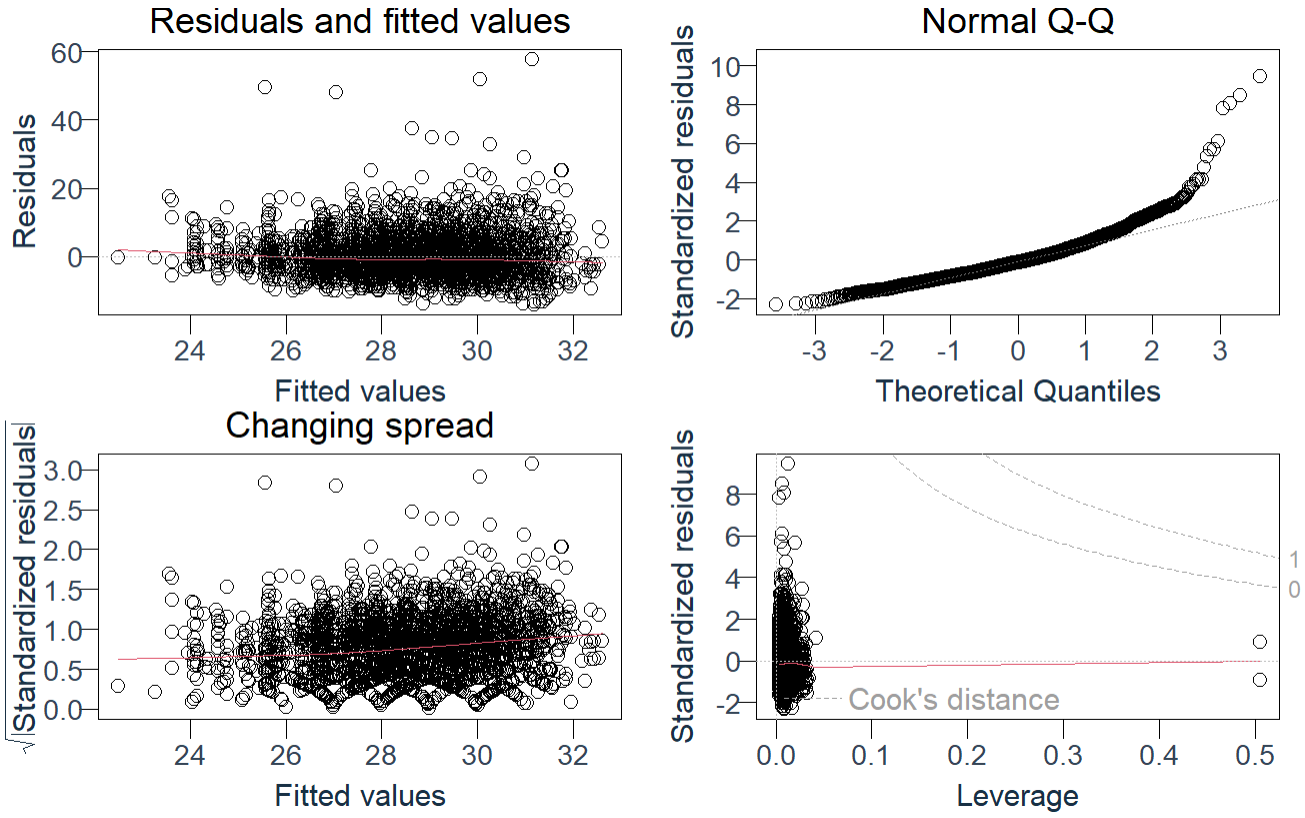
6.81%
R^2

5.99%
Adjusted R^2

Which predictors matter?

Predictor	F	p
Age	9.17	<0.001
Physical activity	13.30	<0.001
Income	4.43	<0.001
Education	8.03	<0.001
Smoking	0.07	0.786

Check the fitted model



Residuals show uneven spread and a long upper tail.

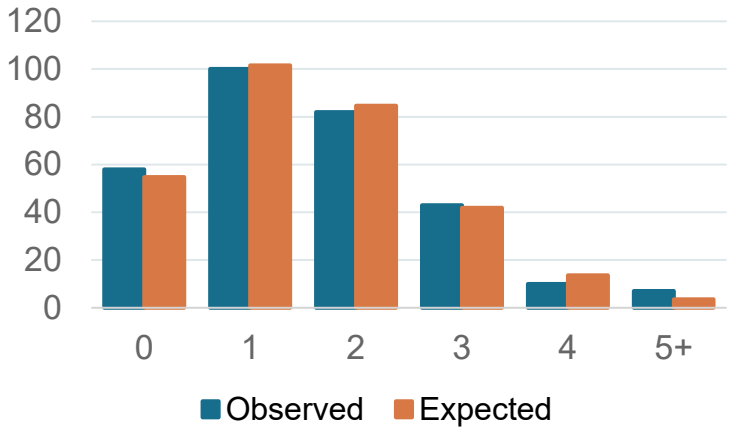
Age, activity, income and education relate to BMI. The model explains only a small share of variation.

Fitting probability distributions

Compare the observed pattern with the pattern each model expects.

Binomial

Positive cases in groups of 10.
p = 0.156, 300 groups.

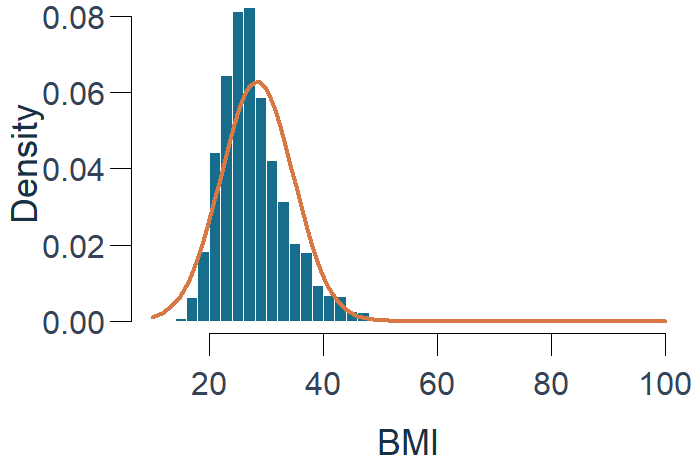


Test p = 0.962

Fits the grouped counts reasonably.

Normal

BMI: mean 28.40, SD 6.35.
The curve misses the long tail.

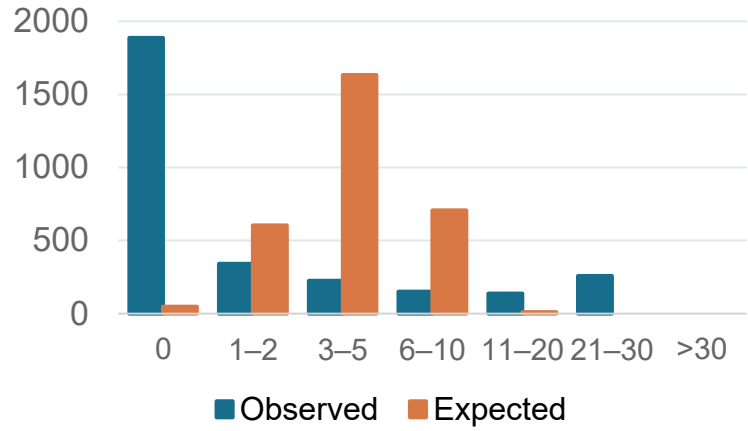


Test p < 0.001

BMI is not normally distributed.

Poisson

Physical-health days: mean 4.15,
variance 74.24. Many zeros.



Test p < 0.001

Variance is far above the mean.

A useful analysis explains why a model fits or fails.

Hypothesis tests and paired measurements

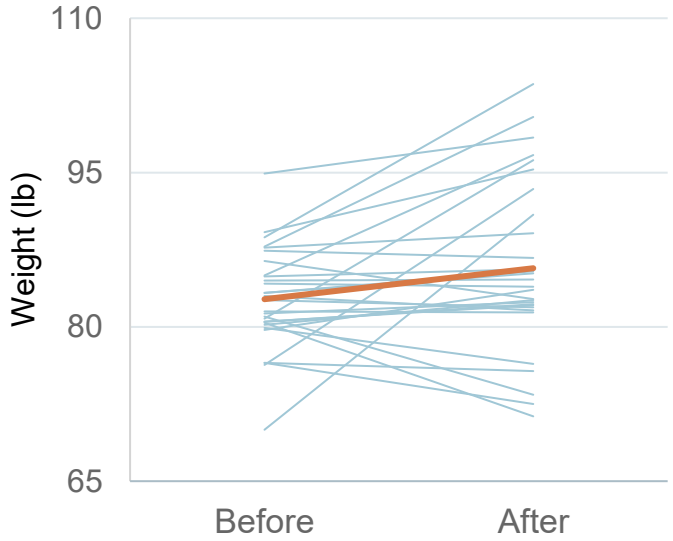
At α = 0.05, p < 0.05 supports a difference. Read the size of the change too.

Six questions, six test results

Question / comparison	Method	Result	p
Mean BMI vs 27	One-sample t	28.40	<0.001
Positive share vs 11%	One proportion	15.63%	<0.001
BMI: female − male	Welch t	−0.65	0.004
Positive share: never / ever smoked	Two proportions	14.09% / 17.36%	0.016
Poor-health days: female − male	Independent t	+1.25 days	<0.001
Weight: after − before	Paired t	+3.01 lb	0.035

Reference values 27 and 11% are classroom assumptions.

Paired example: same 29 patients



Mean gain: +3.01 lb
95% CI: +0.23 to +5.79 lb

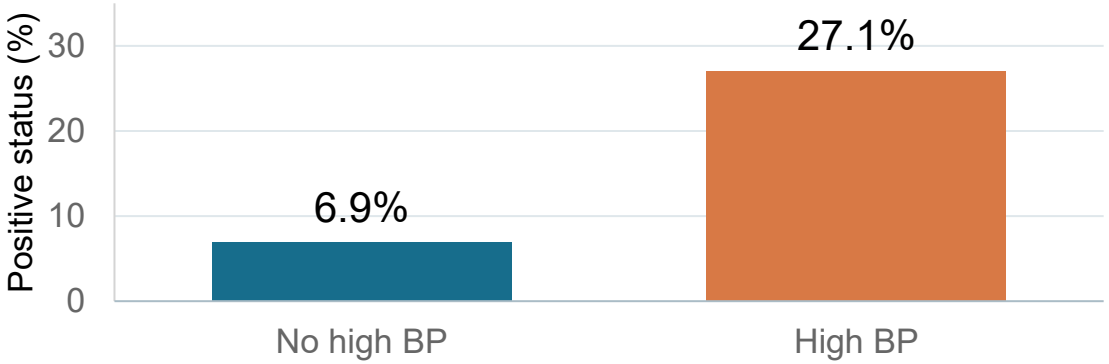
Paired tests compare the same people twice. Independent tests compare different groups.

Chi-square: category patterns and relationships

Goodness of fit checks category counts. Independence tests check whether two categories are related.

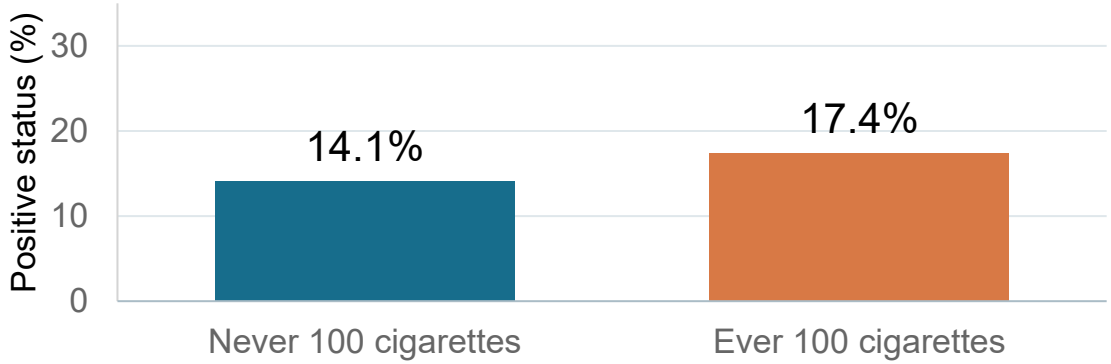
Test	χ^2	p
Age groups: equal frequency (GOF)	678.12	<0.001
Status × high BP	226.28	<0.001
Status × smoking	6.07	0.014

Blood pressure: larger difference



Association size: V = 0.275

Smoking: smaller difference



Association size: V = 0.045

Positive status is more closely related to high blood pressure than to ever-smoking in this sample.

ANOVA and experimental designs

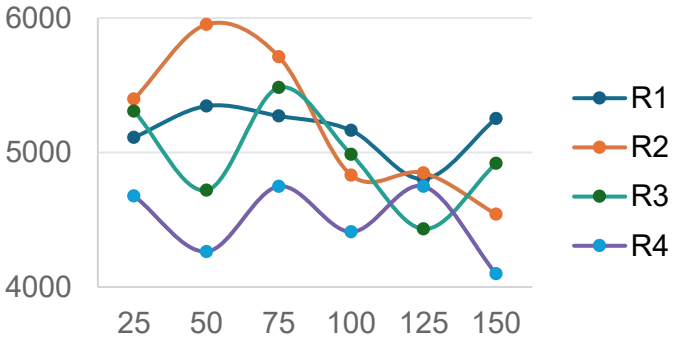
ANOVA compares means. CRD has no blocks. RBD controls one factor and Latin Square controls two.

Comparison	F	p
BMI by education (survey)	14.41	<0.001
Potato CRD: 32 plots, 7 groups	3.61	0.010

Education: existing survey groups.
CRD: 32 randomized plots, 7 sulfur treatments.

Both tests detect a difference between means.
F does not identify every differing pair.

RBD: one blocking factor (rice)



Term	F	p
Block	5.86	0.007
Rate	2.17	0.113

6 rates in each block.
44.8% lower error.

X: seeding rate (kg/ha). Y: yield (kg/ha). Rate p = 0.113.

Latin Square: rows and columns

B	D	E	A	C
C	A	B	E	D
D	C	A	B	E
E	B	C	D	A
A	E	D	C	B

Wheat: 5 treatments in a 5 × 5 grid. Each letter occurs once per row and column.

F = 21.03, p < 0.001

C: highest mean, 13.12. Weekly dusting.

Blocks reduce unexplained error. Weekly dusting (C) has the highest wheat yield in its experiment.

Conclusion: findings and next steps

We described health patterns and applied every course method using documented data.

What we found

Higher average BMI in positive groups

27.72 without diabetes, 32.85 prediabetes, 31.98 diabetes.
Individual values still overlap.

Regression explains a small share

R^2: 1.03% (simple), 6.81% (multiple).
Most BMI variation remains unexplained.

Outliers and model checks matter

Removing 114 flags lowers mean BMI from 28.40 to 27.65.
Binomial fits reasonably. Normal and Poisson fit poorly.

The design changes the comparison

BP has a larger status association than smoking.
Blocks cut error by 44.8%. Wheat treatment C yields most.

Limits

2015, self-reported data.
Small groups add uncertainty.
Associations do not prove causes.
Low R^2 limits prediction.

Next steps

Use newer, better measurements.
Add useful BMI predictors.
Evaluate on unseen data.
Repeat the model checks.

All 11 labs are covered. The findings follow the data and their limitations.